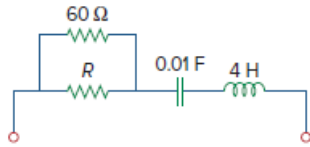


Problem

For the circuit in Fig. 16.38, calculate the value of R needed to have a critically damped response.

Figure 16.38:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

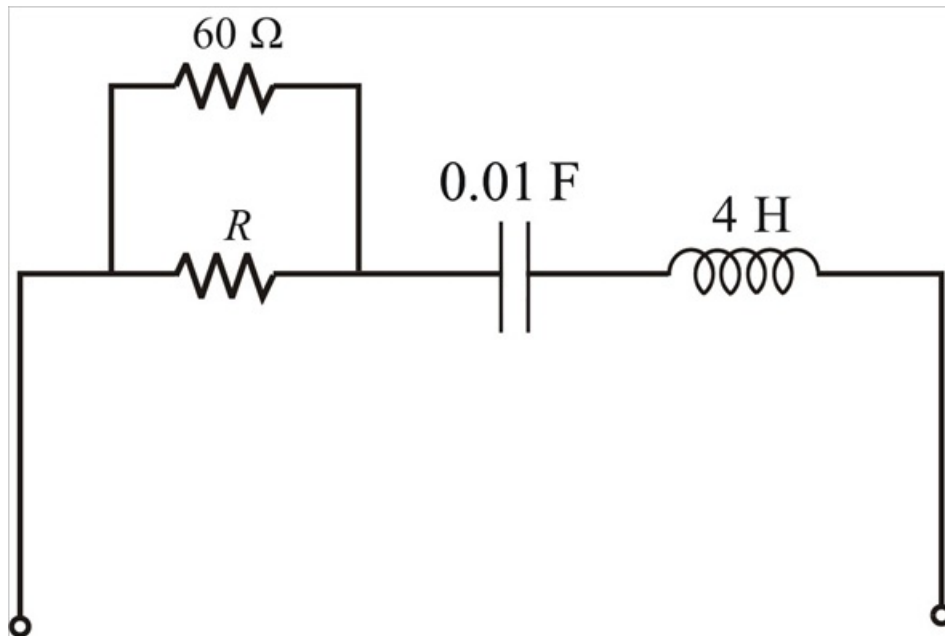


Figure 1

Step 2 of 5

Redraw the circuit in Figure 1, after transforming the circuit in Figure 1.

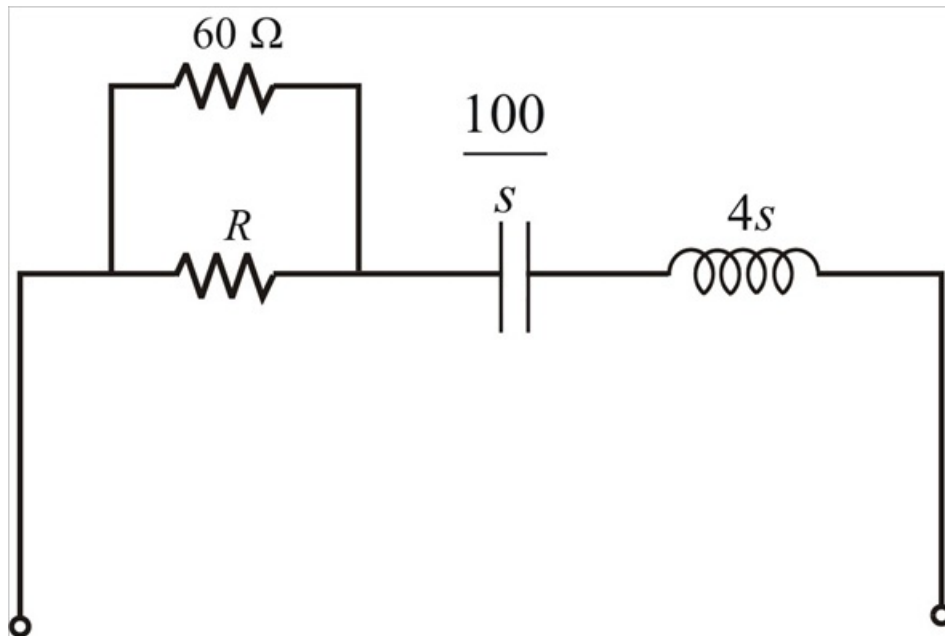


Figure 2

Step 3 of 5

Calculate the impedance of the circuit in Figure 2.

$$\begin{aligned}
 Z &= (R \parallel 60) + \frac{100}{s} + 4s \\
 &= \frac{60R}{60+R} + \frac{100}{s} + 4s \\
 &= \frac{\frac{60R}{60+R}s + 100 + 4s^2}{s} \\
 &= \frac{4}{s} \left[s^2 + \frac{15R}{60+R}s + 25 \right]
 \end{aligned}$$

Step 4 of 5

Characteristic equation of Z .

$$s^2 + \frac{15R}{60+R}s + 25 = 0 \dots\dots (1)$$

Write the general expression for characteristic equation of a series RLC circuit.

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \dots\dots (2)$$

Step 5 of 5

Compare constant terms of the characteristic equations (1) and (2).

$$\omega_n^2 = 25$$

$$\omega_n = 5$$

Compare s coefficients of the characteristic equations (1) and (2).

$$2\zeta\omega_n = \frac{150R}{60+R}$$

$$(2)(1)(5) = \frac{150R}{60+R} \quad (\because \zeta = 1)$$

$$10 = \frac{150R}{60+R}$$

$$600 + 10R = 15R$$

$$R = 120 \, \Omega$$

Thus, the value of the resistor R is $\boxed{120 \, \Omega}$.